Predicting the Replicability of Social Science Lab Experiments

Supplementary Material — January 16, 2019

1 Description of the Data

Explanations of all variables in the data set are available in Table 1. Summary statistics for continuous variables are presented in Table S2, for binary variables in Figure S3 and categorical variables in Figures S4 to S11.

In Table S12 the accuracy of the prediction markets is presented. The ending prices for each stock are directly interpreted as replication probabilities and prediction accuracy is calculated based on a 50% probability cutoff as before.

Table S1: Variable Descriptions

Variable	Description
Avg. Author Citations (O)	Average nr of citation of authors in original study
Avg. Author Citations (R)	Average nr of citation of authors in replication
Max Author Citations (O)	Nr of citation of the author in the original study with the highest citation count
Max Author Citations (R)	Nr of citation of the author in the original study with the highest citation count
Authors (R)	Names of replication authors
Ratio of male authors (O)	Ratio of male authors in original study
Ratio of male authors (R)	Ratio of male authors in replication
Citations (O)	Number of citations of original paper
Compensation (O)	Compensation in original experiment. One of the following: nothing, cash, credit, mixed
Compensation (O): Cash	Subjects were compensated with cash in original experiment
Compensation (O): Credit	Subjects were compensated with course credits in original experiment
Compensation (O): Mixed	Subject were compensated with both cash and credits in original experiment.
Compensation (O): Nothing	No compensation in original experiment
Compensation (R)	Compensation in replication. One of the following: nothing, cash, credit, mixed
Compensation (R): Cash	Subjects were compensated with cash in replication
Compensation (R): Credit	Subjects were compensated with course credits in replication
Compensation (R): Mixed	Subject were compensated with both cash and credits in replication.
Compensation (R): Nothing	No compensation in replication
Discipline	Discipline of original paper. One of the following: social, cognitive, economics.
Discipline: Cognitive Psych.	Cognitive Psychology
Discipline: Economics	Economics
Discipline: Social Psych.	Social Psychology
Effect Size (O)	Standardized effect size of original paper
Effect Size (R)	Standardized effect size of replication
Effect Type	Type of effect tested. One of the following: main effect, correlation, interaction
Effect: Correlation	Experiment tests a correlation effect
Effect: Interaction	Experiment tests an interaction effect
Effect: Main	Experiment tests a main effect
Prediction Market Price	Final market price in prediction market
ES at 80% power	Standardized effect size required in replication to achieve 80% power.
Experiment Country (O)	Country where original experiment was conducted
Experiment Country (R)	Country where replication is to be conducted
Experiment Language (O)	Language used with subjects in original experiment
Experiment Language (R)	Language to be used with subjects in replication
id	Unique id for each O/R pair
Replication Lab ID	Unique id for each replication lab
Paper Length (O)	Number of pages of original paper
Sample Size (O)	Sample size of original paper
Sample Size (R)	Sample size of replication
Number of Authors (O)	Number of authors in original study
Number of Authors (R)	Number of authors in replication
Planned Sample Size (R)	Planned sample size of replication

Online (O)	If the original experiment was conducted online
Online (R)	If the replication was conducted online
P-Value (O)	P-value of original paper
P-Value (R)	P-value of replication
Post-Hoc Power (O)	Post hoc power based on original effect size
Post-Hoc Power (R)	Post hoc power based on replication effect size
Planned Power (R)	Planned power of replication based on planned N and original ES
Replication Project	The replication project that the study was in. One of either: EE, RPP, ML1, ML3
Relative Effect Size	The continuous outcome variable; the standardized replication effect size relative to the original effect.
Replicated	Binary outcome variable, study is replicated if p<=0.05 and effect goes in the same direction as the original.
Same Country	Original study and replication are in the same country
Same Language	Original study and replication are in the same language
Same Online	Original study and replication are both conducted online
Same Subjects	Original study and replication use same type of subjects
Highest Author Seniority (O)	Most senior author in original paper (Professor, Associate Prof., ...)
Seniority (O): Assistant Prof.	Highest seniority in original study is Assistant Professor (rank 3/4)
Seniority (O): Associate Prof.	Highest seniority in original study is Associate Professor (rank 2/4)
Seniority (O): Professor	Highest seniority in original study is Professor (rank 1/4)
Seniority (O): Researcher	Highest seniority in original study is Researcher (rank 4/4)
Highest Author Seniority (R)	Most senior author in original paper (Professor, Associate Prof., ...)
Seniority (R): Assistant Prof.	Highest seniority in replication is Assistant Professor (rank 3/4)
Seniority (R): Associate Prof.	Highest seniority in replication is Associate Professor (rank 2/4)
Seniority (R): Professor	Highest seniority in replication is Professor (rank 1/4)
Seniority (R): Researcher	Highest seniority in replication is Researcher (rank 4/4)
Type of subjects (O)	Type of subjects in original experiment. One of the following: students, community, anyone, online
Type of subjects (R)	Type of subjects in replication. One of the following: students, community, anyone, online
PM Trading Volume	Total volume of traded stocks in prediction market
Number of PM transactions	Number of transactions in prediction market
US Lab (O)	Original experiment lab in the US
US Lab (R)	Replication experiment lab in the US

Table S2: Summary Statistics (continuous variables)

Variable	Min.	Median	Mean	Max.	NA's
author_citations_avg.o	40.00	1733.67	4210.25	44075.00	0.00
author_citations_avg.r	18.50	361.33	754.66	6063.00	0.00
author_citations_max.o	54.00	3266.00	8152.51	45135.00	0.00
author_citations_max.r	35.00	622.00	3320.81	18185.00	0.00
authors_male.o	0.00	0.67	0.71	1.00	0.00
authors_male.r	0.00	0.52	0.56	1.00	0.00
citations	6.00	72.00	264.35	13010.00	0.00
effect_size.o	0.10	0.38	0.41	0.86	0.00
effect_size.r	-0.45	0.15	0.22	0.92	0.00
endprice	0.13	0.68	0.63	0.94	76.00
es_80power	0.04	0.28	0.30	0.95	0.00
length	1.00	10.00	13.17	45.00	0.00
n_authors.o	1.00	3.00	2.83	9.00	0.00
n_authors.r	1.00	2.00	12.00	64.00	0.00
n_planned.r	5.00	100.00	922.87	6336.00	0.00
n.o	7.00	72.00	92.96	660.00	0.00
n.r	7.00	124.00	899.14	6336.00	0.00
p_value.o	0.00	0.01	0.01	0.10	0.00
p_value.r	0.00	0.14	0.26	0.98	0.00
power_planned.r	0.07	0.93	0.86	1.00	0.00
power.o	0.29	0.77	0.76	1.00	0.00
power.r	0.05	0.30	0.48	1.00	0.00
pub_year	1973.00	2008.00	2007.16	2015.00	0.00
relative_es	-0.90	0.43	0.49	2.38	0.00
trading_volume	168.96	777.51	982.58	2849.37	75.00
transactions	28.00	60.00	74.96	193.00	75.00

Table S3: Summary Statistics (binary variables)

Variable	0	1
online.o	129	2
online.r	122	9
replicated	75	56
same_country	67	64
same_language	41	90
same_online	9	122
same_subjects	18	113
us_lab.o	45	86
us_lab.r	53	78

Table S4: Summary Statistics: (discipline)

	Freq
Cognitive	49
Economics	18
Social	64

Table S5: Summary Statistics: (effect_type)

	Freq
correlation	6
interaction	41
main effect	84

Table S6: Summary Statistics: (experiment_country)

Freq	(o)	(r)
Australia	2	1
Austria	1	6
Belgium	1	0
Canada	4	3
France	4	0
Germany	10	16
Hong Kong	0	1
Israel	4	1
Italy	2	3
Netherlands	4	8
Poland	1	0
Singapore	0	5
Spain	1	0
Sweden	0	1
Switzerland	2	0
United Kingdom	9	7
United States	86	78
Uruguay	0	1

Table S7: Summary Statistics: (experiment_language)

Freq	(o)	(r)
Arabic	1	1
Dutch	5	7
English	101	96
France	1	0
French	3	0
German	13	23
Hebrew	3	0
Italian	2	3
Polish	1	0
Spanish	1	1

Table S8: Summary Statistics: (compensation)

Freq	(o)	(r)
cash	47	52
credit	57	58
mixed	18	15
nothing	9	6

Table S9: Summary Statistics: (subjects)

Freq	(o)	(r)
anyone	9	6
community	8	8
online	0	4
students	114	113

Table S10: Summary Statistics: (project)

Freq	
ee	18
ml1	13
ml3	10
rpp	90

Table S11: Summary Statistics: (seniority)

Freq	(o)	(r)
Assistant	0	14
Assistant Professor	2	22
Associate Professor	11	24
Professor	113	57
Researcher	5	14

Table S12: Prediction Market accuracy (at 50% probability cutoff)

	Full dataset	ML dataset
Pooled PM Accuracy:	66.7%	65.5%
RPP PM Accuracy:	69.2%	67.6%
EE PM Accuracy:	61.1%	61.1%

1.1 Data management and decisions

For most variables we abstain from any sort of imputation, dropping the observation if we were not able to collect all data. However for cases when the planned sample size has not been imported, and thus the planned power cannot be calculated, we do estimate it using actual replication sample size as a proxy.

Since the ML data is pooled, whenever there is a lab-specific variation we use the modal option. If one study was replicated by three French, one Italian and two American labs, we mark the study as having been conducted in France.

Last, while the publication year of the original article is an ex ante relevant correlate of reproducibility, we have chosen to remove it from the training data since it also captures a feature of the paper selection process used in the ML projects. The authors wanted a benchmark for comparison, and included a number of papers that had been successfully replicated many times before. These studies are all older, making the publication year variable a proxy for papers with (already known) high replication rates.

Citations for papers and authors changes constantly. We downloaded citation counts in September 2016. These could easily be updated in future versions of the model.

Post-hoc power was calculated by the authors based on the hypothesis tests, effect sizes, and sample sizes used in the original experiment.

1.2 Correlations

Figures S1 and S2 show two ways to analyze the correlational structure of the data. The first is a plot of Spearman correlations between all pairs of variables, the full version of the plot also shown in the paper. The second shows the two most important principal components based on the continuous features of the data. In Table S13 the contribution of all 15 components is presented. As one can see, contributed variance tapers off quite slowly, indicating lack of linear structure. In the plot, it seems like the second component captures differences in statistical variables such as p-value and effect size, while the first explains author statistics.

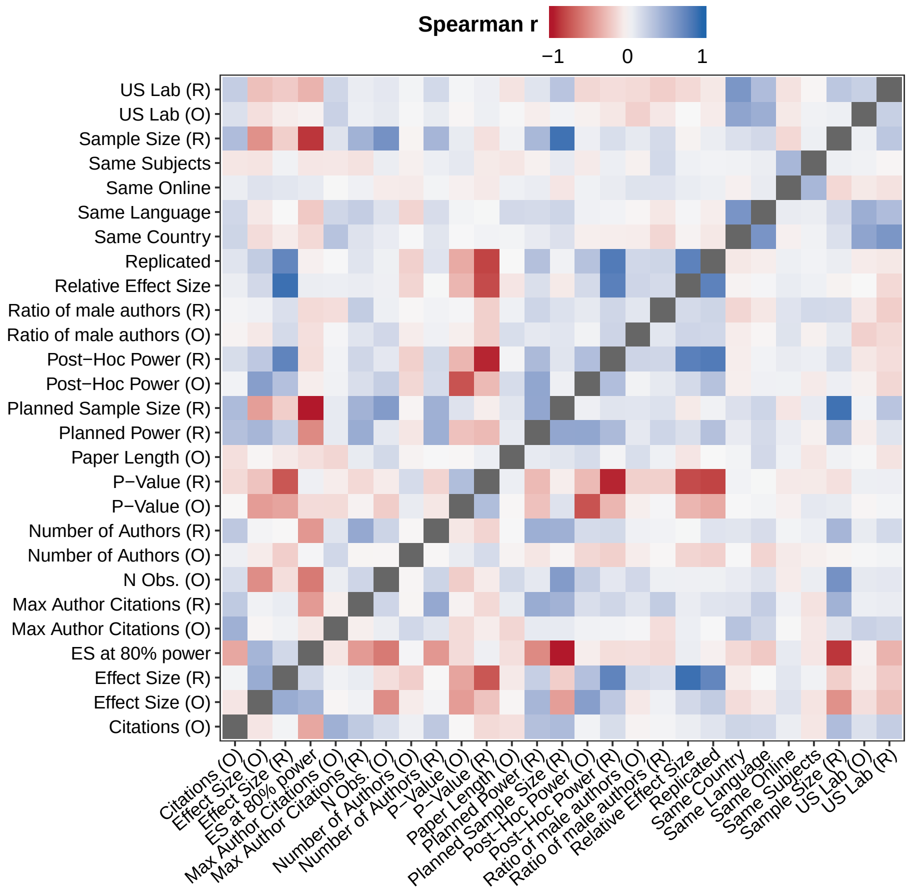

Figure S1: Full Correlation Plot

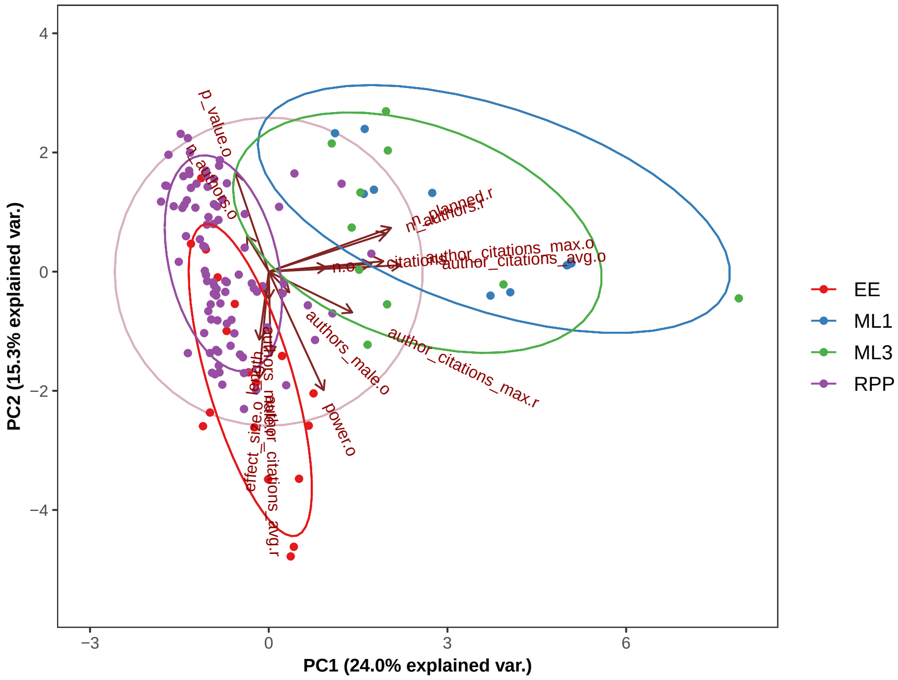

Figure S2: Two most important principal components that together explain 40% of the variation in the continuous features of the data.

	s.d.	Var. \%	Cumulative
PC1	1.898	0.240	0.240
PC2	1.515	0.153	0.393
PC3	1.436	0.137	0.531
PC4	1.185	0.094	0.624
PC5	1.031	0.071	0.695
PC6	1.005	0.067	0.762
PC7	0.932	0.058	0.820
PC8	0.900	0.054	0.874
PC9	0.825	0.045	0.920
PC10	0.732	0.036	0.956
PC11	0.615	0.025	0.981
PC12	0.407	0.011	0.992
PC13	0.282	0.005	0.997
PC14	0.176	0.002	0.999
PC15	0.109	0.001	1.000

Table S13: PCA

2 Training and Model Selection

2.1 Resampling through nested CV

Because we want to (1) compare different models, and (2) estimate model accuracy, we employ the nested cross validation procedure that is described in the paper.

First, we use the procedure to compare a set of different algorithms and models, choosing to analyze the performance of the Random Forest algorithm trained on the whole feature set more in-depth as it performed the best. The results of these CV runs is shown in Figure S3. For each model, we tune its hyperparameters to maximize the AUC. For Random Forest, we evaluate how many variable to randomly sample as candidates at each split and use a fixed number of trees (1001). For LASSO we tune the shrinkage parameter λ, with fix σ at 0.02 and let the cost parameter C vary. Last, for GBM, we vary the number of splits performed in each tree.

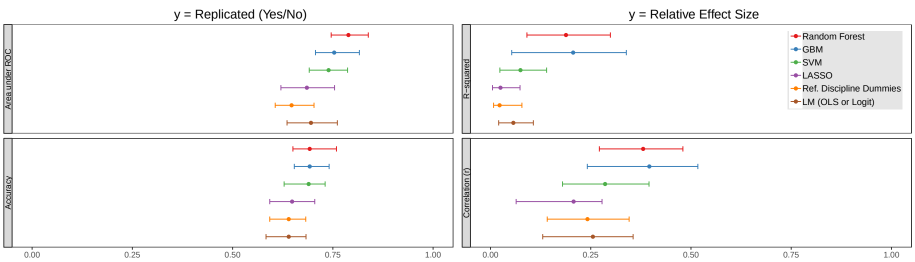

Figure S3: IQR and median for different classification (left) and regression (right) models.

3 Out of Sample Validation

Figure S4 contains predictions and outcomes based on the first stage of the data collection only. The SSRP had a two-stage design. For replications that were found to be not significant in an initial test, more data was collected. Sample sizes are thus smaller for those studies that continued to a second data collection. The model is not very good at capturing this mechanical increase in replication probability from increase sample size. For one thing, we do not have any within-study varition in sample size in the data. While we could use the disaggregated Many Labs data to study within study variation, the sample sizes do not differ much between labs.

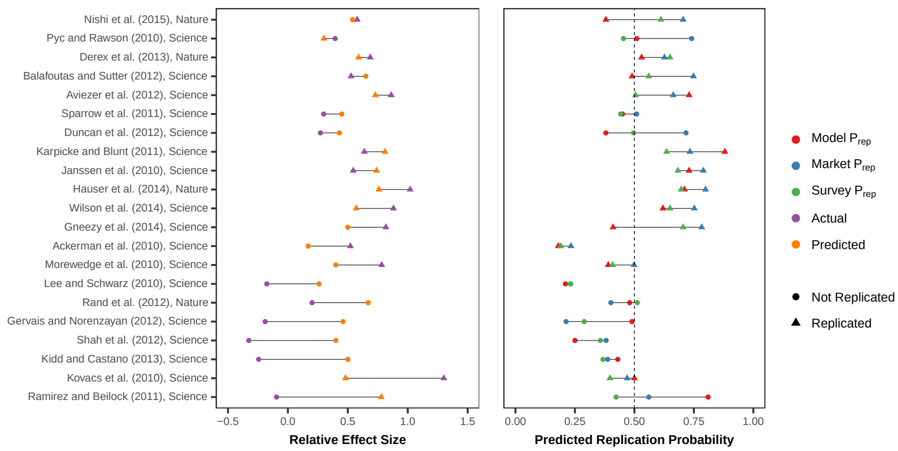

Figure S4: Predicted and actual results of the SSRP. Based on data and predictions from Stage 1 only.

4 Theoretical upper bound of classification model

Statistical tests are designed to make conclusions from noisy data, but this also necessarily implies that such conclusions will sometimes be wrong. Predicting the outcome of these conclusions, if the noise is completely random, can thus never become perfect. If the statistical test has 90% power for example, one out of 10 true effects will fail to replicate. Similarly, a significance level of 5% will mean that one in 20 null effects will still be deemed significant. To see this more clearly, consider the following to tables:

Table S14: Replication Outcomes and Model Predictions

		Outcome of replication	
		Not Replicate	Replicate
Prediction	Not Replicate	True Negative	False Negative
	Replicate	False Positive	True Positive

Table S15: Replication Outcomes and the true state

		Outcome of replication	
		Not Replicate	Replicate
True State	H_0 True	Correct, $1-\alpha$	Type I error, α
	H_0 False	Type II Error, β	Power, $1-\beta$

A prediction can be correct or not in different ways, as becomes clear in Table S14. However, even when the prediction is correct, it does not necessarily mean that the actual effect exists in the population. Instead, it could have been generated by the inherent noisiness of sampled data through Type I and Type II errors. When running replications, we think of these errors as bad scientific luck, and their presence is a reason why multiple replications by independent labs are helpful. But for our purposes, by definition, they make perfect predictions impossible.

Assuming that the noise we see is completely unpredictable, the best a perfect model could do is to always predict replication success when the null is false, and a failure otherwise. Such a model is what we think should be considered the upper bound to which our results can be compared.

So how should we estimate this bound? A perfect model that always predicts the true state would still have an error rate of $1 -$ Power for true effects and $\alpha = 5\%$ for null effects. To know the accuracy of a perfect model, we thus need the proportion of true effects to null effects in our set of studies, and also the power of all tests of true effects. Most replications in our data set aimed at being able to detect the effect of the original study with 90% power. But since effect sizes are almost always smaller than what was initially claimed, real power lies below 90% in most of the replications. With a significance rate of 5%, while we do not know the true proportion of null effects, even a rather conservative measure would put the true upper bound on accuracy somewhere below 90%.